

<i>Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.</i>						
Team Members	Role/ Responsibility	Email				
IZA Prince Boaz	Data preparation, PCA core implementation, GitHub repo	b.iza@alustudent.com				
Bior Aguer Kuir	Variance analysis, visualization, benchmarking, written report	b.aguerkui@alustudent.com				
Task Allocation and Tracker						
	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Download EICV5 household microdata from NISR	IZA Prince Boaz	27/09/2026	Completed	Yes	Dataset meets requirements: African data, 7+ columns, missing values, non-numeric columns
	Convert data to CSV and select PCA features	IZA Prince Boaz	27/09/2026	Completed	Yes	Extracted 11 household economic and demographic columns; converted to clean CSV
	Clean missing values, encode categories, prepare data	IZA Prince Boaz	28/09/2026	Completed	Yes	Data handling completed with missing value imputation and encoding using NumPy exclusively
	Standardize data, compute covariance matrix, write notes	IZA Prince Boaz	28/09/2026	Completed	Yes	Features standardized and covariance matrix computed; covariance explanation completed
	Compute eigendecomposition and sort eigenvalues/eigenvectors	IZA Prince Boaz	29/09/2026	Completed	Yes	Eigenvalues and eigenvectors computed and sorted in descending order; verified
	Calculate explained variance and select components	Bior Aguer Kuir	29/09/2026	Completed	Yes	Explained and cumulative variance calculated; component trade-off explanation documented
	Project data onto PCs, plot results, write analysis	Bior Aguer Kuir	29/09/2026	Completed	Yes	Data projected onto principal components; before and after plots and interpretations generated

	Optimize PCA implementation and benchmark runtime	Bior Aguer Kuir	30/09/2026	Completed	Yes	Vectorized PCA optimization completed; runtime benchmarked and analyzed
	Finalize GitHub repo, export PDFs, and submit	IZA Prince Boaz	30/09/2026	Completed	Yes	Final notebook, data, and documentation committed; repo: https://github.com/boaziza/formative2-pca-pair-team-34

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
2026-09-27	Project kickoff, dataset selection, and task delegation	IZA Prince Boaz	IZA Prince Boaz, Bior Aguer Kuir	Reviewed requirements, selected EICV5 dataset, split tasks	Download microdata, export CSV, begin data prep	
2026-09-28	Data preprocessing and covariance calculation review	Bior Aguer Kuir	IZA Prince Boaz, Bior Aguer Kuir	Verified feature selection, imputation logic, and covariance formula	Finalize covariance script and start eigendecomposition	
2026-09-29	Eigendecomposition and component selection strategy	IZA Prince Boaz	IZA Prince Boaz, Bior Aguer Kuir	Confirmed eigenvalues sort order, variance threshold, and elbow plot	Code variance plot, projection functions, and explanations	
2026-09-30	PCA projection visualization and runtime optimization	Bior Aguer Kuir	IZA Prince Boaz, Bior Aguer Kuir	Reviewed PCA scatter plots and loop vs vectorized benchmarks	Run final benchmarks and document information retention/loss	
2026-09-30	Final peer review, code audit, and Canvas submission prep	IZA Prince Boaz	IZA Prince Boaz, Bior Aguer Kuir	Audited notebook for pure-NumPy compliance and reviewed repo	Push final commits and submit deliverables on Canvas	